

Module 6: Inner Product Spaces

Definition: Let $V(F)$ be a vector space where F is a field of real or complex numbers. An inner product on V is a function $f: V \times V \rightarrow F$ such that

- (i) $f(\alpha, \beta) = \overline{f(\beta, \alpha)}$, where $\overline{f(\beta, \alpha)}$ is the complex conjugate of $f(\beta, \alpha)$, for all $\alpha, \beta \in V$
- (ii) $f(\alpha, \alpha) > 0$ for $\alpha \neq \bar{0}$ and $f(\alpha, \alpha) = 0$ for $\alpha = \bar{0}$
- and (iii) $f(a\alpha + b\beta, r) = af(\alpha, r) + bf(\beta, r)$ for all $\alpha, \beta, r \in V$ and $a, b \in F$.

The vector space $V(F)$ in which an inner product f defined as above is called an inner product space and it is denoted by (V, f) .

Notation: In general $f(\alpha, \beta)$ is denoted by $\langle \alpha, \beta \rangle$ or (α, β) or $\langle \alpha | \beta \rangle$. Here after we use $\langle \alpha, \beta \rangle$ for $f(\alpha, \beta)$.

Therefore, the above conditions of the inner product are written as follows:

- (i) $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$
- (ii) $\langle \alpha, \alpha \rangle > 0$ for $\alpha \neq \bar{0}$ and $\langle \alpha, \alpha \rangle = 0$ for $\alpha = \bar{0}$.
- and (iii) $\langle a\alpha + b\beta, r \rangle = a\langle \alpha, r \rangle + b\langle \beta, r \rangle$
for all $\alpha, \beta, r \in V, \bar{0} \in V$
and $a, b, \bar{0} \in F$.

Example:

① If $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$ are the elements of a vector space $\mathbb{R}^3(\mathbb{R})$, then $\langle \alpha, \beta \rangle = x_1 x_2 + y_1 y_2 + z_1 z_2$ is an inner product on $\mathbb{R}^3$.

Note: The above product is called the dot product of α and β and is denoted by $\alpha \cdot \beta$.

$$\text{i.e., } \langle \alpha, \beta \rangle = \alpha \cdot \beta.$$

This is called the standard inner product on $\mathbb{R}^3$.

② For $A, B \in M_{n \times n}(\mathbb{R})$, define $\langle A, B \rangle = \text{tr}(B^T A)$ is an inner product on $M_{n \times n}(\mathbb{R})$.

Note: $\text{tr}(B^T A) = \text{trace of } (B^T A)$

Norm or Length of a vector:

Definition: Let $V(F)$ be an inner product space.

The norm (or, length) of a vector $\alpha \in V$ is denoted by $\|\alpha\|$ and defined as the positive

square root of $\langle \alpha, \alpha \rangle$.

$$\text{i.e., } \|\alpha\| = \sqrt{\langle \alpha, \alpha \rangle}$$

Note: ① For $\alpha \in V$, $\|\alpha\|^2 = \langle \alpha, \alpha \rangle$

② In an inner product space $V(F)$,
(i) $\|a\alpha\| = |a| \|\alpha\|$ for $\alpha \in V$ and $a \in F$.

(ii) $|\langle \alpha, \beta \rangle| \leq \|\alpha\| \|\beta\|$ for $\alpha, \beta \in V$.

(iii) $\|\alpha + \beta\| \leq \|\alpha\| + \|\beta\|$ for $\alpha, \beta \in V$

Definition: Let $V(F)$ be an inner product space.

A vector $\alpha \in V$ is said to be a unit vector if $\|\alpha\| = 1$.

Definition: Let $V(F)$ be an inner product space and $\alpha, \beta \in V$. α is said to be orthogonal to β if $\langle \alpha, \beta \rangle = 0$.

Orthogonal and orthonormal sets

Definition: Let $V(F)$ be an inner product space.

A non-empty sub set S of V is said to be

(i) an orthogonal if for every pair of distinct vectors of S are orthogonal.

$$\text{i.e., for every } \alpha_i, \alpha_j \in S, \langle \alpha_i, \alpha_j \rangle = 0 \quad (i \neq j),$$

i.e., for every α_i, α_j ($i \neq j$) in S , $\langle \alpha_i, \alpha_j \rangle = 0$.

(ii) an orthonormal if S is orthogonal and for every $\alpha \in S$, $\|\alpha\| = 1$.

Examples:

① $S = \left\{ \left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3} \right), \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right), \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right) \right\}$ is an orthonormal set in $\mathbb{R}^3(\mathbb{R})$.

② $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthonormal set in $\mathbb{R}^3(\mathbb{R})$.

③ $S = \{(1, -2, -2), (2, -1, 2), (2, 2, -1)\}$ is an orthogonal set in $\mathbb{R}^3(\mathbb{R})$ but it is not an orthonormal set.

Gram-Schmidt orthogonalization Process

Let $V(F)$ be an inner product space and $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a basis for V . Let $\beta_1 = \alpha_1$,

$$\beta_2 = \alpha_2 - \frac{\langle \alpha_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1$$

$$\beta_3 = \alpha_3 - \frac{\langle \alpha_3, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle \alpha_3, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2$$

$\vdots$

$$\beta_n = \alpha_n - \frac{\langle \alpha_n, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \dots - \frac{\langle \alpha_n, \beta_{n-1} \rangle}{\langle \beta_{n-1}, \beta_{n-1} \rangle} \beta_{n-1}$$

Then $\{\beta_1, \beta_2, \dots, \beta_n\}$ is an orthogonal basis for V .

And hence $\left\{ \frac{\beta_1}{\|\beta_1\|}, \frac{\beta_2}{\|\beta_2\|}, \dots, \frac{\beta_n}{\|\beta_n\|} \right\}$ is an orthonormal basis for V .

Note: Every finite dimensional inner product space has an orthogonal basis and ~~thus~~ hence it has an orthonormal basis.

Problems:

① Given $\{(2, 1, 3), (1, 2, 3), (1, 1, 1)\}$ is a basis of $\mathbb{R}^3(\mathbb{R})$. Obtain an orthogonal basis of $\mathbb{R}^3(\mathbb{R})$ using Gram Schmidt orthogonal process.

Sol: Let $\alpha_1 = (2, 1, 3)$, $\alpha_2 = (1, 2, 3)$, $\alpha_3 = (1, 1, 1)$.

By Gram Schmidt orthogonalization process an orthogonal basis $\{\beta_1, \beta_2, \beta_3\}$ is given by

$$\beta_1 = \alpha_1 = (2, 1, 3).$$

$$\beta_2 = \alpha_2 - \frac{\langle \alpha_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 = (1, 2, 3) - \frac{\langle (1, 2, 3), (2, 1, 3) \rangle}{\langle (2, 1, 3), (2, 1, 3) \rangle} (2, 1, 3)$$

$$= (1, 2, 3) - \frac{13}{14} (2, 1, 3)$$

$$= \frac{1}{14} (-12, 15, 3)$$

$$\beta_3 = \alpha_3 - \frac{\langle \alpha_3, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle \alpha_3, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2$$

$$= (1, 1, 1) - \frac{\langle (1, 1, 1), (2, 1, 3) \rangle}{14} (2, 1, 3)$$

$$- \frac{\langle (1, 1, 1), \frac{1}{14}(-12, 15, 3) \rangle}{\left(\frac{378}{196}\right)} \frac{1}{14}(-12, 15, 3)$$

$$= (1, 1, 1) - \frac{6}{14} (2, 1, 3) - \frac{\frac{6}{14} \cdot \frac{1}{14} (-12, 15, 3)}{\left(\frac{378}{196}\right)}$$

$$= (1, 1, 1) - \frac{6}{14} (2, 1, 3) - \frac{6}{378} (-12, 15, 3)$$

$$= (1, 1, 1) - \frac{6}{14} (2, 1, 3) - \frac{1}{378} (-72, 90, 18)$$

$$= \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}\right)$$

Therefore, $\left\{ (2, 1, 3), \left(-\frac{12}{14}, \frac{15}{14}, \frac{3}{14}\right), \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}\right) \right\}$

is an orthogonal basis of $\mathbb{R}^3(\mathbb{R})$.

Hence $\left\{ \left(\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right), \left(\frac{-12}{\sqrt{378}}, \frac{15}{\sqrt{378}}, \frac{3}{\sqrt{378}}\right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) \right\}$
is an orthonormal basis of $\mathbb{R}^3(\mathbb{R})$.

- ② Use Gram Schmidt process to obtain an orthogonal basis of $\mathbb{R}^3(\mathbb{R})$, given that $\{(1, 0, 1), (1, 0, -1), (0, 3, 4)\}$ is a basis of $\mathbb{R}^3(\mathbb{R})$ and hence find an orthonormal basis of $\mathbb{R}^3(\mathbb{R})$

Answer: $\left\{ \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right), (0, 1, 0) \right\}$
is an orthonormal basis of $\mathbb{R}^3(\mathbb{R})$

- ③ Prove that $\left\{ \left(\frac{3}{5}, 0, \frac{4}{5} \right), \left(-\frac{4}{5}, 0, \frac{3}{5} \right), (0, 1, 0) \right\}$ is an orthonormal set in $\mathbb{R}^3$.

- ④ on $P_2(\mathbb{R})$, consider an inner product as

$$\langle p(x), q(x) \rangle = \int_0^1 p(x) \cdot q(x) dx$$

Apply Gram-Schmidt Process to the basis $\{1, x, x^2\}$ of P_2 to produce an orthogonal basis of $P_2(\mathbb{R})$.

Sol: Taking $\alpha_1 = 1, \alpha_2 = x, \alpha_3 = x^2$

Now, $\beta_1 = \alpha_1 = 1$

$$\begin{aligned} \beta_2 &= \alpha_2 - \frac{\langle \alpha_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 = x - \frac{\langle x, 1 \rangle (1)}{\langle 1, 1 \rangle} = x - \frac{\int_0^1 (x)(1) dx}{\int_0^1 (1)(1) dx} \\ &= x - \frac{1}{2} \end{aligned}$$

$$\beta_3 = \alpha_3 - \frac{\langle \alpha_3, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle \alpha_3, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2 = \dots$$

Practice all the questions

1. If $\alpha = (1, 3, -1)$, $\beta = (-1, 1, 2)$ are two vectors in an inner product space $\mathbb{R}^3(\mathbb{R})$,

find (i) $\|\alpha\|$ (ii) $\langle \alpha, \beta \rangle$ (iii) angle between α and β .

Answer: (i) $\|\alpha\| = \sqrt{\langle \alpha, \alpha \rangle} = \sqrt{11}$ (ii) $\langle \alpha, \beta \rangle = 1(-1) + 3(1) + (-1)2 = 0$

(iii) $\cos \theta = \frac{\langle \alpha, \beta \rangle}{\|\alpha\| \|\beta\|} = \frac{0}{\sqrt{11}\sqrt{6}} = 0 \Rightarrow \theta = \frac{\pi}{2}.$

2. Is $S = \{(1, -1, 1), (0, 1, 1), (2, 1, -1)\}$ an orthogonal set in an inner product space $\mathbb{R}^3(\mathbb{R})$.

Answer: Clearly every pair of distinct vector are orthogonal. Hence S is an orthogonal set in $\mathbb{R}^3(\mathbb{R})$.

3. Is $S = \left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right) \right\}$ an orthonormal set in an inner product space $\mathbb{R}^2(\mathbb{R})$.

Answer: Clearly S is an orthogonal set and each vector in S is a unit vector. Hence S is an orthonormal set in $\mathbb{R}^2(\mathbb{R})$.

4. Using Gram Schmidt process finds an orthogonal basis for $\mathbb{R}^3(\mathbb{R})$, given that

$\{(1, 2, 2), (-1, 0, 2), (0, 0, 1)\}$ is a basis for $\mathbb{R}^3$. Also find an orthonormal basis for $\mathbb{R}^3(\mathbb{R})$.

Answer: $\{(1, 2, 2), (-4/3, -2/3, 4/3), (2/9, -2/9, 1/9)\}$ is an orthogonal basis for $\mathbb{R}^3(\mathbb{R})$

and hence its orthonormal basis is $\left\{ \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right), \left(-\frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right), \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right) \right\}.$

5. On $P_2(\mathbb{R})$, consider an inner product as $\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx$. Apply Gram-Schmidt

process to the basis $\{1, x, x^2\}$ to produce an orthogonal basis of $P_2(\mathbb{R})$ and hence find an orthonormal basis.